

Problem

Design a problem to help other students better understand how to use *PSpice* or *MultiSim* to solve circuit problems with periodic inputs.

Step-by-step solution

Step 1 of 1

Design a problem to help other students better understand how to use *PSpice* to solve circuit problems with periodic inputs.

Although there are many ways to solve this problem, this is an example based on the same kind of problem asked in the third edition.

Problem

Rework Prob. 17.40 using *PSpice*.

Chapter 17, Problem 40.

The signal in Fig. 17.77(a) is applied to the circuit in Fig. 17.77(b). Find $v_o(t)$.

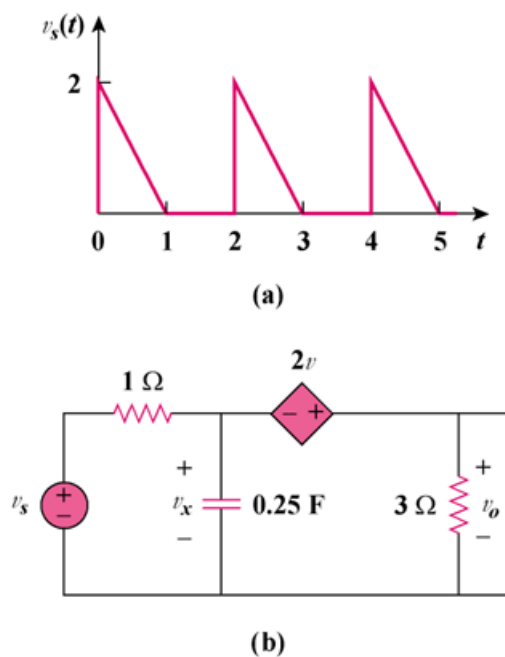
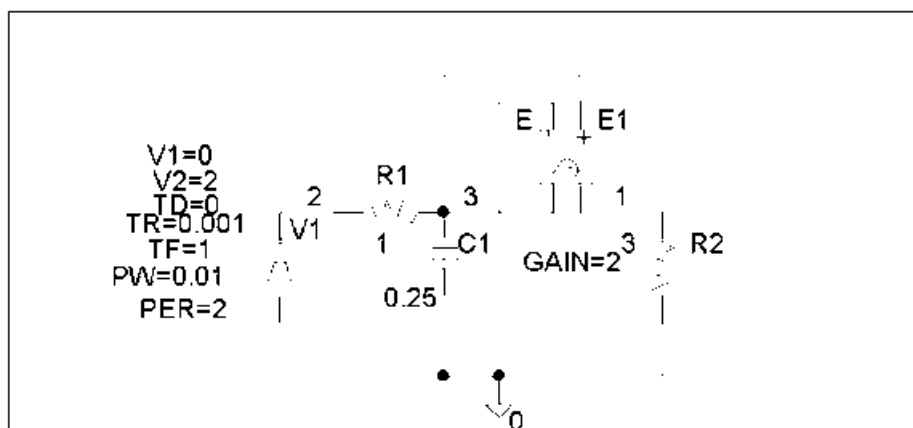
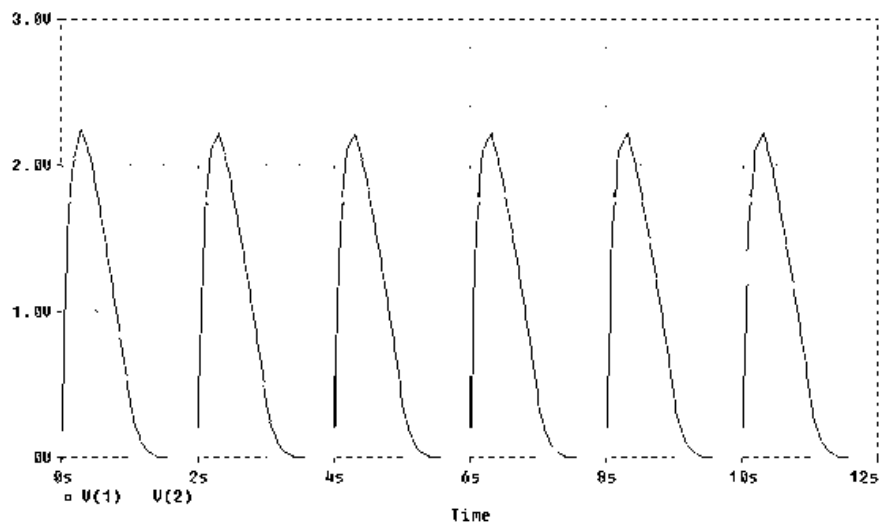


Figure 17.77

Solution

The schematic is shown below. In the Transient dialog box, we set Print Step = 0.02 s, Final Step = 12 s, Center Frequency = 0.5, Output Vars = V(1) and V(2), and click enable Fourier. After simulation, we compare the output and output waveforms as shown. The output includes the following Fourier components.





FOURIER COMPONENTS OF TRANSIENT RESPONSE V(1)

DC COMPONENT = 7.658051E-01

HARMONIC	FREQUENCY	FOURIER	NORMALIZED	PHASE	NORMALIZED
NO	(HZ)	COMPONENT	COMPONENT	(DEG)	PHASE(DEG)
1	5.000E-01	1.070E+00	1.000E+00	1.004E+01	0.000E+00
2	1.000E+00	3.758E-0	3.512E-01	-3.924E+01	-4.928E+01
3	1.500E+00	2.111E-01	1.973E-01	-3.985E+01	-4.990E+01
4	2.000E+00	1.247E-01	1.166E-01	-5.870E+01	-6.874E+01
5	2.500E+00	8.538E-02	7.980E-02	-5.680E+01	-6.685E+01
6	3.000E+00	6.139E-02	5.738E-02	-6.563E+01	-7.567E+01
7	3.500E+00	4.743E-02	4.433E-02	-6.520E+01	-7.524E+01
8	4.000E+00	3.711E-02	3.469E-02	-7.222E+01	-8.226E+01
9	4.500E+00	2.997E-02	2.802E-02	-7.088E+01	-8.092E+01

TOTAL HARMONIC DISTORTION = 4.352895E+01 PERCENT

